

TB_EQ User Manual

A spectral dynamic equalizer with Basic EQ, Dynamic EQ, and Spectral Resonance Reduction in one band workflow

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TB_EQ - Spectral Dynamic Equalizer

1. Overview

TB_EQ is an equalizer that starts with a conventional parametric EQ band and optionally adds Dynamic EQ or Spectral resonance reduction. Each band always has a Basic EQ shape. Dynamic and Spectral processing are additional layers, and only one of them can be active on a band at a time.

Default behavior: a new plugin instance opens with Band 1 enabled and selected. Its default setting is Bell, 1000 Hz, 0 dB, Q 1.0, so it does not change the sound but keeps the interface populated.

Main features

Feature	Description
Basic EQ	Up to 24 bands. Supports Bell, Low Shelf, High Shelf, Low Cut, High Cut, Notch, and Band Pass.
Dynamic EQ	Per-band Threshold, Range, Attack, and Release control the amount of EQ movement.
Spectral EQ	STFT-based resonance detection reduces sharp resonant components with Sensitivity and Range controls.
Audition	Hold right-click to monitor the band area or, in Spectral mode, the resonance component being reduced.
Analyzer	Shows spectrum, Basic EQ response, Dynamic/Spectral movement, and expected ranges.

2. Quick Start

Task	How to do it
Select a band	Click a node or choose a band from the Band selector.
Create a band	Double-click an empty area in the analyzer.
Delete a band	Double-click a band node. Newly created bands are briefly protected from accidental immediate deletion.
Edit Frequency/Gain	Drag a node. Left-right changes Frequency, up-down changes Gain.
Edit Q	Use the mouse wheel over the node.
Reset a control	Double-click a knob, or Alt-click it.

3. Interface

The main analyzer displays the live spectrum and the EQ response. The vertical scale is centered around 0 dB and spans +24 dB to -24 dB. The green curve is the Basic EQ response. Band nodes show the frequency and gain position of each active band.

Element	Meaning
Green curve	Basic EQ response.
Band node	Frequency and Gain position of a band.
Orange area/line	Dynamic EQ expected range and actual movement.
Purple area/line	Spectral resonance reduction movement.
INPUT/OUTPUT	Fixed horizontal sliders in the lower-right area. They do not move when a band mode changes.

4. Bands and Basic EQ

Every band has a Basic EQ shape. Enabling Dynamic or Spectral mode keeps that Basic EQ shape active and adds the selected processing layer. Dynamic and Spectral modes are mutually exclusive on the same band.

Control	Description
Type	Select Bell, Low Shelf, High Shelf, Low Cut, High Cut, Notch, or Band Pass.
Freq	Center or reference frequency. It uses a logarithmic feel for finer low-frequency control.
Gain	Boost or cut amount for Bell and Shelf types.
Q	Bandwidth for Bell, Notch, and Band Pass. It also shapes the transition feel for Shelf and Cut types.
Enabled	Turns the selected band on or off.

5. Dynamic EQ

Dynamic mode moves the selected EQ band according to signal level. Negative Range values reduce the band, while positive Range values boost it. The analyzer shows the full expected movement range as a translucent area and the current movement as a solid line.

Control	Description
Threshold	The level where movement begins.
Range	Maximum amount of movement. Negative values cut, positive values boost.
Attack	How quickly the movement reacts when the signal crosses the threshold.
Release	How quickly the movement returns when the signal falls back.
Tip: For vocal harshness or low-end boom, use a negative Range and lower the Threshold until the movement line appears only when needed.	

6. Spectral EQ

Spectral mode uses STFT analysis to find narrow resonant components that stand out from nearby frequencies. Its reduction line is drawn relative to the Basic EQ curve, so you can see how much additional resonance reduction is applied after the Bell or Shelf shape.

Control	Description
Sensitivity	0.0 to 1.0. Higher values catch subtler resonances and allow stronger reduction.
Range	The frequency span affected by spectral processing. It behaves like a broad/narrow range control.
Q	Shapes the Basic EQ width and contributes to the perceived focus of the Spectral band.
Current implementation: with high Sensitivity, Spectral reduction can reach up to -24 dB. If it feels too strong, lower Sensitivity or narrow the Range.	

7. Right-click Audition

Mode	What you hear
Basic or Dynamic band	Hold right-click to monitor the selected band through a Band Pass audition filter.
Spectral band	Hold right-click to hear the resonant component that Spectral processing is trying to reduce.
Right-click drag	While auditioning, drag left-right for Frequency and up-down for Gain.

Mode	What you hear
Mouse wheel while auditioning	Adjusts the selected band's Q.

8. Input/Output Gain and Reset

Input Gain and Output Gain are fixed horizontal sliders in the lower-right section. They stay in place regardless of the selected band mode. Knobs and sliders can be reset with double-click or Alt-click.

9. Troubleshooting

Symptom	What to check
The plugin opens with no visible band	Make sure the latest build is loaded. This version enables Band 1 by default.
A newly created band disappears	This version adds protection against immediate accidental deletion. If it still happens, the DAW may be loading an old VST3.
The Spectral line appears detached from the EQ curve	Rescan or reload the latest VST3. The current drawing logic attaches Spectral movement to the Basic EQ curve.
The VST3 does not appear in the DAW	Check the Debug build output folder and the DAW's VST3 search path.